STA 325 Case Study Report

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Introduction

Predicting when inertial particles form large, concentrated clusters can help researchers understand and anticipate uneven particle distributions in turbulent flows, including flows relevant to clouds. But resolving each new condition with direct numerical simulation can take days or longer. A reliable statistical surrogate could therefore help researchers screen conditions, prioritize costly simulations, and explore how changes in turbulence, gravity, or particle inertia reshape clustering. In 89 simulations, our collaborators identified clusters and recorded their volume distributions at different combinations of three parameters:

- **Reynolds number** Re: 90, 224, 398. This measures how strong the turbulence is.
- **Froude number** Fr: 0.052, 0.3, ∞. This measures the strength of gravity. Smaller Fr means stronger gravity, and $Fr = \infty$ means no gravity.
- **Stokes number** St: 14 values between 0.05 and 3. This measures particle inertia. Particles with small St follow the flow closely, while particles with large St are heavier and respond more slowly.

We pursue two connected goals. First, we predict the mean, standard deviation, skewness, and kurtosis of cluster volume at new (Re, Fr, St) settings. These summaries describe typical cluster size and the frequency of unusually small or large clusters, helping collaborators compare conditions before committing to another expensive simulation. Second, we infer from our model how turbulence, gravity, and inertia shape each summary: which effects are connected with the response, and whether one parameter's influence changes with another. This inference can point to the conditions that promote concentrated clusters and guide which regimes merit closer physical study.

For prediction, we use response-specific GAMs with a shared smooth of $\log St$ and an $Re \times Fr$ interaction: $k = 5$ for mean and standard deviation, and $k = 4$ for skewness and kurtosis, as selected by LOOCV.

$$\log(responses) = \beta_0 + \beta_1 f_{smooth}(\log(St)) + \beta_2 Re + \beta_3 Fr + \beta_4 Re : Fr + \epsilon.$$

Methodology

Responses and transformations

Each run reports the first four raw moments of its cluster-volume distribution, which we converted to central moments with `raw2central` and then to the mean, standard deviation, skewness, and

Table 1: LOOCV MSE (log scale) for each candidate model. For the GAM, the parameter count is its effective degrees of freedom for the mean.

Model	Parameters	Mean	Standard deviation	Skewness	Kurtosis
Baseline (additive)	6	0.0192	0.7338	0.3561	1.3252
Interaction Model 1	13	0.0104	0.3290	0.0715	0.2119
Interaction Model 2	9	0.0114	0.2565	0.0554	0.1656
Cubic Model	11	0.0093	0.1377	0.0480	0.1567
Natural spline (4 df)	12	0.0090	0.1447	0.0484	0.1569
GAM (k = 4)	10.9	0.0090	0.1451	**0.0478**	**0.1541**
GAM (k = 5)	11.7	**0.0089**	**0.1356**	0.0484	0.1559

kurtosis. The four responses differ in skewness: the mean and standard deviation are strongly right-skewed, while skewness and kurtosis are only mildly skewed, so a single log transform might not suit all four. We therefore fit a separate Box-Cox transformation to each response. Every 95% interval for λ contains 0, the log transform (Table 2), so the data support logging all four responses. We also log-transformed St, which is right-skewed and spans a 60-fold range. Re and Fr each take only 3 values, so we treat them as categorical.

Model selection

We compared the linear candidates and two GAM specifications by leave-one-out cross-validation (LOOCV), holding out one run at a time and measuring prediction error on the log-response scale (Table 1). The linear candidates include additive and interaction models, a cubic polynomial in $\log St$, and a natural spline with 4 degrees of freedom. The natural spline and cubic model both allow a nonlinear St effect while retaining the $Re \times Fr$ interaction. Each GAM uses the same $Re \times Fr$ terms and a shared smooth of $\log St$; its smoothness is estimated by REML. The basis dimension k sets the maximum flexibility of the smooth, and we compared $k = 4$ and $k = 5$.

The lowest LOOCV error depends on the response: the $k = 5$ GAM performs best for mean and standard deviation, while the $k = 4$ GAM performs best for skewness and kurtosis. We therefore use the corresponding GAM for each response rather than selecting one specification for all four. The bold entries in Table 1 identify each response's lowest-error candidate. The GAM parameter counts are effective degrees of freedom (EDF) for the mean-response fit; the EDF can differ by response. For inference, we use the cubic model, which describes the St effect with a closed-form curve and standard linear-model confidence intervals. Its inferential conclusions also agree with the GAMs'. We use the best GAM for each prediction task.

Diagnostics

The selected GAMs fit the log responses closely, with R^2 from 0.972 to 0.999. Residual-versus-fitted plots show that the selected GAMs remove the broad group-level offsets seen under the additive baseline, with residuals centered around zero across the (Re, Fr) combinations (Figure 3 in the Appendix). Some uneven residual spread and a few large errors remain, especially for standard deviation, skewness, and kurtosis. Runs sharing (Re, Fr) may come from the same simulated flow and therefore may not be independent; this could make uncertainty intervals too narrow.

Results

Predictions on the hold-out set

We predicted the four statistics at the 23 hold-out settings (all point predictions are in `predictions.csv`). Because each GAM models the log of a statistic, exponentiating its fitted value gives the predicted median on the original scale. Every hold-out setting lies within the training range, so no prediction requires extrapolation, although the two settings at $St = 0.05$ sit at the edge of the design.

Effects of each parameter

Because the model is fit on the log scale, exponentiated contrasts are multiplicative changes, and because Re and Fr interact, the effect of gravity depends on Re. Figure 1 shows the fitted value of each statistic for every (Re, Fr) combination.

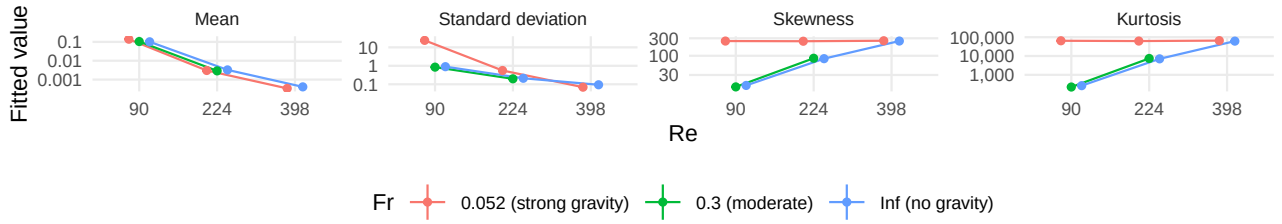


Figure 1: Fitted values at St = 1 for each (Re, Fr) combination, with 95% confidence intervals (log scale). Lines connect the same gravity level across turbulence levels; there are no runs at Re = 398, Fr = 0.3.

Turbulence intensity (Re) controls cluster size. Under strong gravity, the cubic model estimates that increasing Re from 90 to 398 shrinks mean cluster volume by about 410 times and standard deviation by about 340 times. With no gravity, the reductions are about 250 times for the mean and 9.6 times for standard deviation. Stronger turbulence produces a wider range of eddy sizes, including much smaller ones, which appears to break particles into much smaller clusters.

Strong gravity creates heavy tails, but only when turbulence is weak. At $Re = 90$, moving from strong gravity ($Fr = 0.052$) to no gravity ($Fr = \infty$) reduces skewness by a factor of about 16 and kurtosis by about 244, so strong gravity is associated with many small clusters and occasional very large ones. This effect weakens as Re increases: removing gravity reduces kurtosis by about a factor of 9 at $Re = 224$ and has little effect at $Re = 398$, visible in Figure 1 as a gap at $Re = 90$ that closes as Re increases. One physical interpretation is that as turbulence strengthens, turbulent velocity fluctuations overwhelm gravitational settling, so the flow rather than gravity determines how particles cluster.

Moderate gravity behaves like no gravity. The fitted values for $Fr = 0.3$ and $Fr = \infty$ are similar across the four responses, so within the range studied, gravity is associated with clustering mainly when it is strong. Because there are no runs at $Fr = 0.3$ with $Re = 398$, this comparison covers only the two lower turbulence levels.

Particle inertia (St) has a smaller, nonlinear effect. Figure 2 shows the effect of St relative to $St = 1$. Mean cluster volume rises with St, the standard deviation rises steeply at low St and then

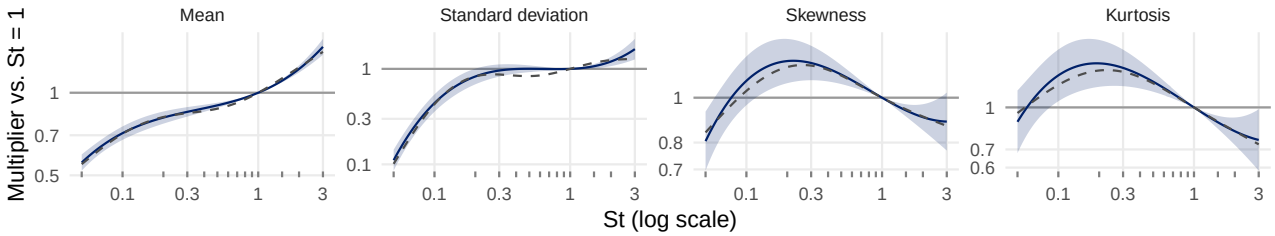


Figure 2: Estimated effect of St on each statistic, as a multiplier relative to St = 1, from the cubic model (solid line, 95% confidence band) and the tuned GAM (dashed line). Tick marks show the observed St values.

levels off, and skewness and kurtosis peak at intermediate St before declining. The cubic and GAM curves agree closely, so these conclusions do not depend on the polynomial form, and the bands widen near $St = 0.05$, where the model fits least well. Because the slope changes across the range, no single "doubling St" percentage summarizes the effect.

Conclusion

Each parameter shapes the cluster-size distribution in a different way. Turbulence intensity is strongly associated with cluster scale, while strong gravity is associated with heavier tails when turbulence is weak. The effect of particle inertia is smaller and changes across the observed St range. Because gravity's influence varies with turbulence strength, conclusions drawn at one turbulence level should not be assumed to hold at another.

Limitations and future work

The model can only predict at the 3 simulated levels of Re and Fr: because it treats them as categorical, it cannot interpolate between levels or extrapolate beyond them, and snapping a new input to the nearest level would ignore the difference between levels. Future simulations should therefore be spread off the grid. The current grid spends all 89 runs on 3 levels each of Re and Fr, and refining it scales badly: adding a fourth level of each parameter raises a full grid from $3^3 = 27$ to $4^3 = 64$ settings. A space-filling design gives every run new values of all three parameters, so the same compute budget covers far more of the input space and lets Re and Fr be treated as continuous.

Appendix

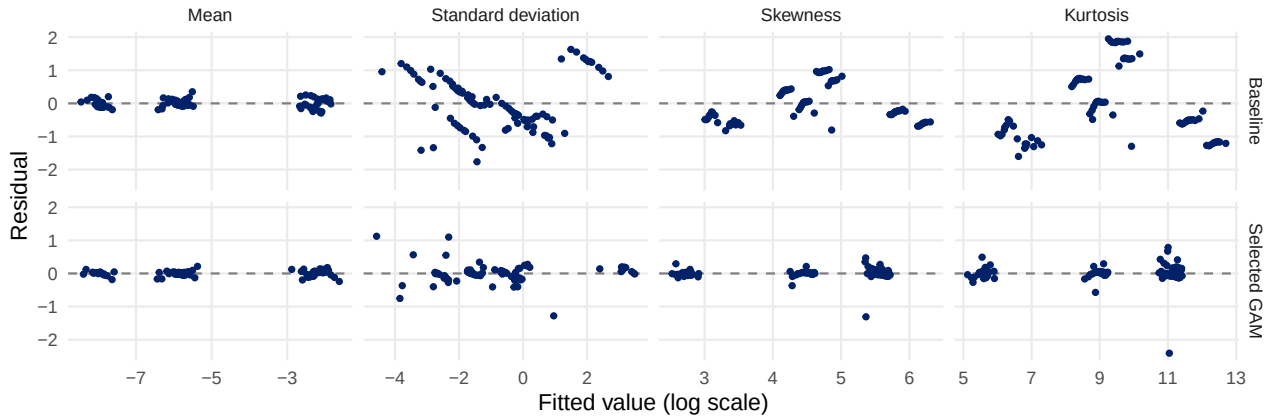


Figure 3: Residuals against fitted values for the additive baseline (top) and the response-specific selected GAMs (bottom).

Table 2: Estimated Box-Cox λ for each response, with 95% profile-likelihood intervals. $\lambda = 0$ is the log transform.

Response	Estimate	95% interval
Mean	-0.02	[-0.05, 0.01]
Standard deviation	-0.03	[-0.07, 0.01]
Skewness	-0.11	[-0.28, 0.07]
Kurtosis	-0.05	[-0.12, 0.02]